

Lecture #10 / October 6th, 2026

NO CLASS NEXT TUESDAY (Monday schedule)

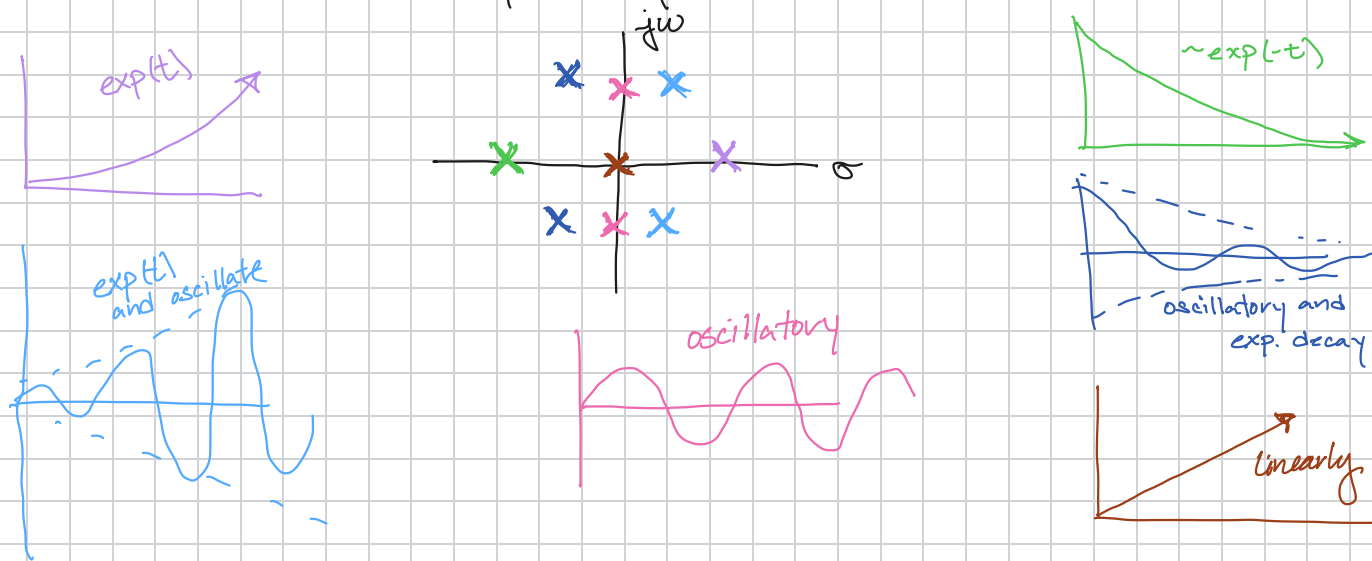
HW #2 due ~~Thursday~~: ~~electronically by 11:59 PM~~ ~~in-person by 3:30 PM~~ ~~either option works~~

Thursday Oct. 15 All Problems 1-5

AI use is okay for "hints" on getting started, but NOT okay for solving the problem for you.

- Time-domain Specs $\rightarrow$ "Performance" (How quickly?)
- Stability $\rightarrow$ "Convergence" (Does it?)

Q: Where do we place our poles in order achieve both stability and performance?



The gist: in order to get stable performance, we have to place our poles in the Left Half Plane

Q: How do we place our poles in the LHP?

Answer: Feedback Control

Last Time: Routh's Criterion $\rightarrow$ a tool for determining whether a polynomial has any roots in the RHP (and if so, how many?)

Polynomials $\rightarrow$ Characteristic Polynomials $\rightarrow$ Pole Locations

$$a(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

- ① every coefficient of $a(s)$ is present (i.e., nonzero) and positive.
- ② Define the Routh Array

$$\begin{array}{c|ccc} s^n & 1 & a_2 & a_4 & \dots \\ s^{n-1} & a_1 & a_3 & a_5 & \dots \\ s^{n-2} & b_1 & b_2 & b_3 & \\ s^{n-3} & c_1 & c_2 & c_3 & \\ \vdots & \vdots & \vdots & \vdots & \\ 1 & & & & \end{array}$$

$$b_1 = -\frac{1}{a_1} \det \begin{bmatrix} 1 & a_2 \\ a_1 & a_3 \end{bmatrix}$$

$$b_2 = -\frac{1}{a_1} \det \begin{bmatrix} 1 & a_4 \\ a_1 & a_5 \end{bmatrix}$$

$$c_1 = -\frac{1}{b_1} \det \begin{bmatrix} a_1 & a_3 \\ b_1 & b_2 \end{bmatrix}$$

$\vdots$

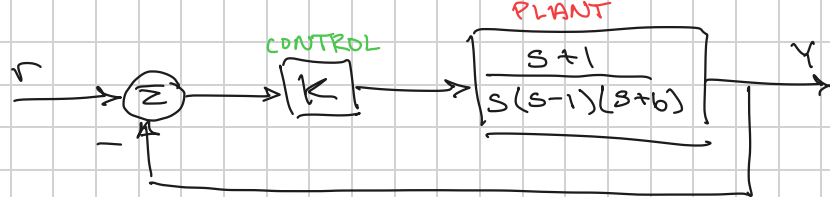
All roots are strictly in the LHP if the column is positive.
The number of sign changes is the number of RHP poles.

Q: Why do we care? "roots" exists...

A: Routh Array is symbolic, but "roots" is numeric.

$\rightarrow$ Allows us to analyze feedback loops as a function of control (plant) parameters.

Example:



Control "K" is just a gain (proportional control)

What we want to do: determine what values of "K"

stabilize the closed-loop system.

- 1) Find the CL characteristic polynomial (as a function of "K")
- 2) Use Routh Array to determine set of inequalities that determine what values of "K" make char. poly. stable.

① $T = \frac{Y}{R} = \frac{KP}{1+KP} \Rightarrow s(s-1)(s+b) + K(s+1) = 0$
(characteristic polynomial)
denominator of this goes to zero

Expand CP $\Rightarrow s^3 + ss^2 + (K-b)s + K = 0$

② what values of "K" make the roots of CP in the LHP?

① $K-b > 0$ AND $K > 0$
 $(K > b)$

②

s^3	1	$K-b$	
s^2	5	K	
s^1	$\frac{4K-30}{5}$	?	} Don't Care
s^0	K	?	

> 0

We learned that $\frac{4K-30}{5} > 0 \Rightarrow 4K > 30$

$K > 30/4$

- ① $K > 0$ (don't care) ② $K > b$ (don't care) ③ $K > 7.5$ (care)

Let's assume that $K > 0$ but $K < 7.5$

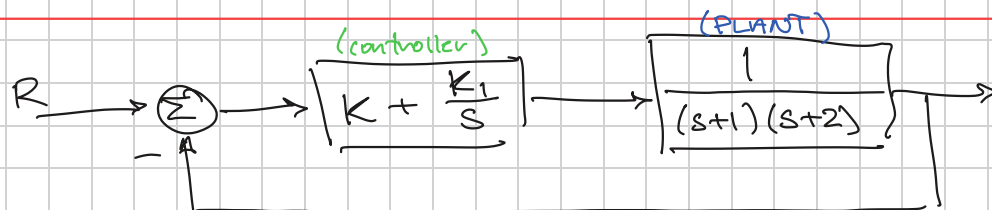
s^3	1	← +	$\Rightarrow 2$ sign changes $\Rightarrow 2$ poles in the RHP
s^2	5	← +	
s^1	$\frac{4K-30}{5}$	← -	
s^0	K	← +	

Right Half Plane

Let's assume $K < 0$

s^3	1	← +	$\Rightarrow 1$ sign change $\Rightarrow 1$ pole in the right half plane
s^2	5	← +	
s^1	$\frac{4K-30}{5}$	← -	
s^0	K	← -	

Example



Goal:

Controller has 2 parameters, and we want to determine what values of K and K_1 lead to a stable Ch system.

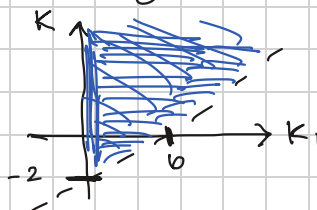
① Char. Poly. $T = \frac{Y}{R} = \frac{KP}{1+KP} \Rightarrow s(s+1)(s+2) + Ks + K_1 = 0$

$\Rightarrow s^3 + 3s^2 + (2+K)s + K_1 = 0$

② Routh Criterion: $K > -2$ AND $K_1 > 0$ (all coeffs positive)

s^3	1	$2+K$
s^2	3	K_1
s^1	$\frac{6+3K-K_1}{3}$	?
s^0	K_1	?

$s_0: \frac{6+3K-K_1}{3} > 0 \Rightarrow K > \frac{K_1}{3} - 2$



$$\begin{array}{l} s^3 \\ s^2 \\ s^1 \\ s^0 \end{array} \left| \begin{array}{lll} (a_0) & 1 & (a_2) 2+k & (a_4) \dots \\ (a_1) & 3 & (a_3) k_1 & (a_5) \dots \\ & b_1 & & \end{array} \right.$$

$$b_1 = -\frac{1}{a_1} \det \begin{bmatrix} 1 & a_2 \\ a_1 & a_3 \end{bmatrix}$$

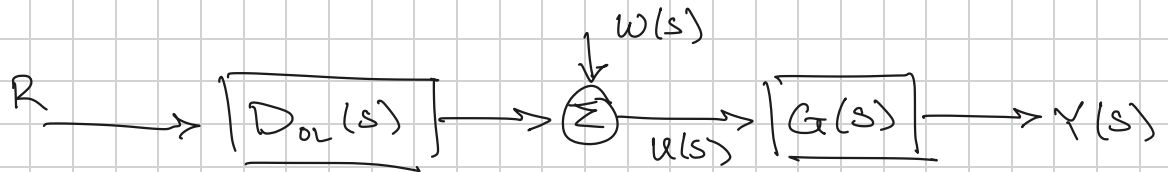
$$= -\frac{1}{3} \det \begin{bmatrix} 1 & 2+k \\ 3 & k_1 \end{bmatrix}$$

$$= -\frac{1}{3} (k_1 - 3(2+k))$$

$$= -\frac{1}{3} (k_1 - 6 - 3k)$$

We have mostly focused on $T(s) = \frac{KP}{1+KP} = \frac{Y}{R}$

①
OPEN LOOP



R is the reference

$G(s)$ is plant

D_{OL} is the controller

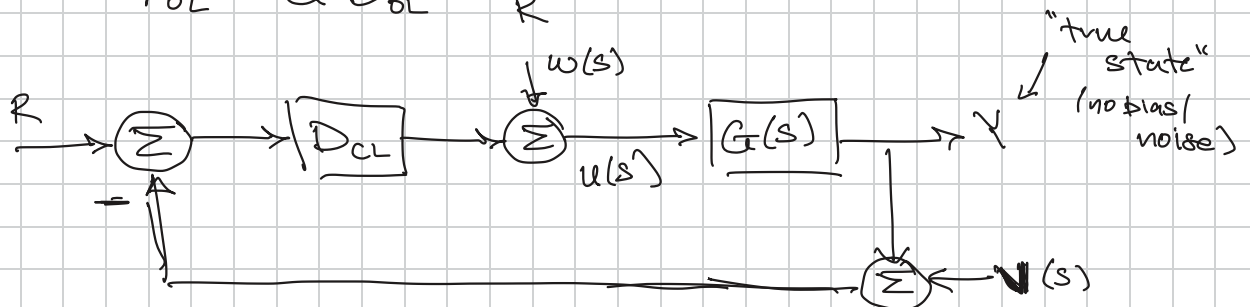
$w(s)$ is a disturbance

$$Y = G D_{OL} R + G w$$

$$E_{OL} = R - Y = (1 - G D_{OL}) R - G w$$

$$T_{OL} = G D_{OL} = \frac{Y}{R}$$

②
CLOSED LOOP



D_{CL} is the closed loop controller

$v(s)$ is measurement noise / disturbance (sensor-related)

$w(s)$ is (still) the disturbance

Track: Y needs to "track" $R(s)$

Regulate: Y needs to "reject" $W(s)$

Filter: Y needs to "attenuate" $V(s)$

} CL system has to perform all of these

(Plant's output) $Y = \frac{GD_{cl}}{1+GD_{cl}} R + \frac{G}{1+GD_{cl}} W - \frac{GD_{cl}}{1+GD_{cl}} V$

Plant's input (actuation) $U = \frac{D_{cl}}{1+GD_{cl}} R - \frac{GD_{cl}}{1+GD_{cl}} W - \frac{D_{cl}}{1+GD_{cl}} V$ (noise sensitivity)

Tracking Error ($R-Y$) $E_{cl} = \frac{1}{1+GD_{cl}} R - \frac{G}{1+GD_{cl}} W + \frac{GD_{cl}}{1+GD_{cl}} V$ (load sensitivity)

Book refers to these as the "Gang of Four"

$T = \frac{GD_{cl}}{1+GD_{cl}}$
(complementary sensitivity function)

$S = \frac{1}{1+GD_{cl}}$
(sensitivity function)

$T + S = 1$

Takeaway #1) Open Loop systems cannot place poles

whatever your plant's dynamics are, you are stuck with, because you cannot modify them with feedback

Closed loop systems can place poles

Pole: $T(s) = \frac{N(s)}{D(s)}$

Zeros are $N(s)=0$
Poles are $D(s)=0$

Example:

$$G(s) = \frac{1}{s^2 + 3s + 9}$$

$$D_{CL} = \frac{c_2 s^2 + c_1 s + c_0}{s(s + d_1)}$$

Req: Need char poly to be the following:

$$(s+b)(s+3)(s^2+3s+9) = 0 \quad (\text{time-domain spec's})$$

What are (c_1, c_2, c_0, d_1) that give the char. polynomial

$$\textcircled{1} \quad S = \frac{1}{\underbrace{1 + D_{CL}G}_{=0}}$$

$$\Rightarrow 1 + D_{CL}G = 0$$

$$1 + \left(\frac{c_2 s^2 + c_1 s + c_0}{s(s + d_1)} \right) \left(\frac{1}{s^2 + 3s + 9} \right) = 0$$

Work through next time.

$$c_2 = 18, \quad c_1 = 54, \quad c_0 = 162, \quad d_1 = 9$$